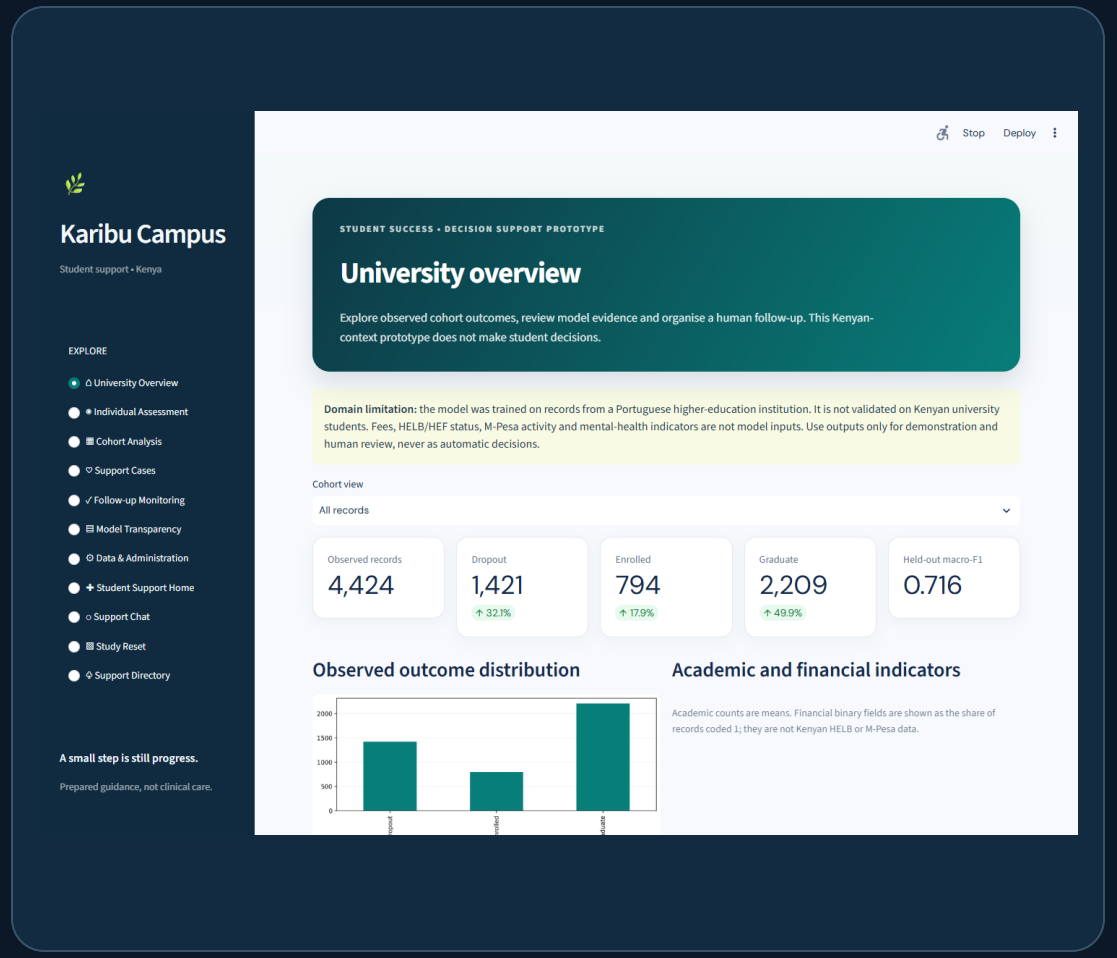
KCA UNIVERSITY | CPP 4201

KARIBU CAMPUS

Student Support and Academic Outcome Prototype

Accessible student support alongside analysis of historical academic outcomes.

Advanced Python Programming | Project Presentation



University Overview · source cohort and held-out metric

The problem and our approach

Student pressures overlap, while support routes and outcome insights can be difficult to bring together.

CAMPUS CHALLENGES

- CATs, exams, coursework and disengagement
- Fees, HELB or HEF uncertainty, upkeep and transport
- Finding a practical, trusted support route
- Learning from patterns in historical academic outcomes

TWO COMPLEMENTARY COMPONENTS

Student Support Hub

Student Support Home · Intent-based Chat
Study Reset planner · Support Directory

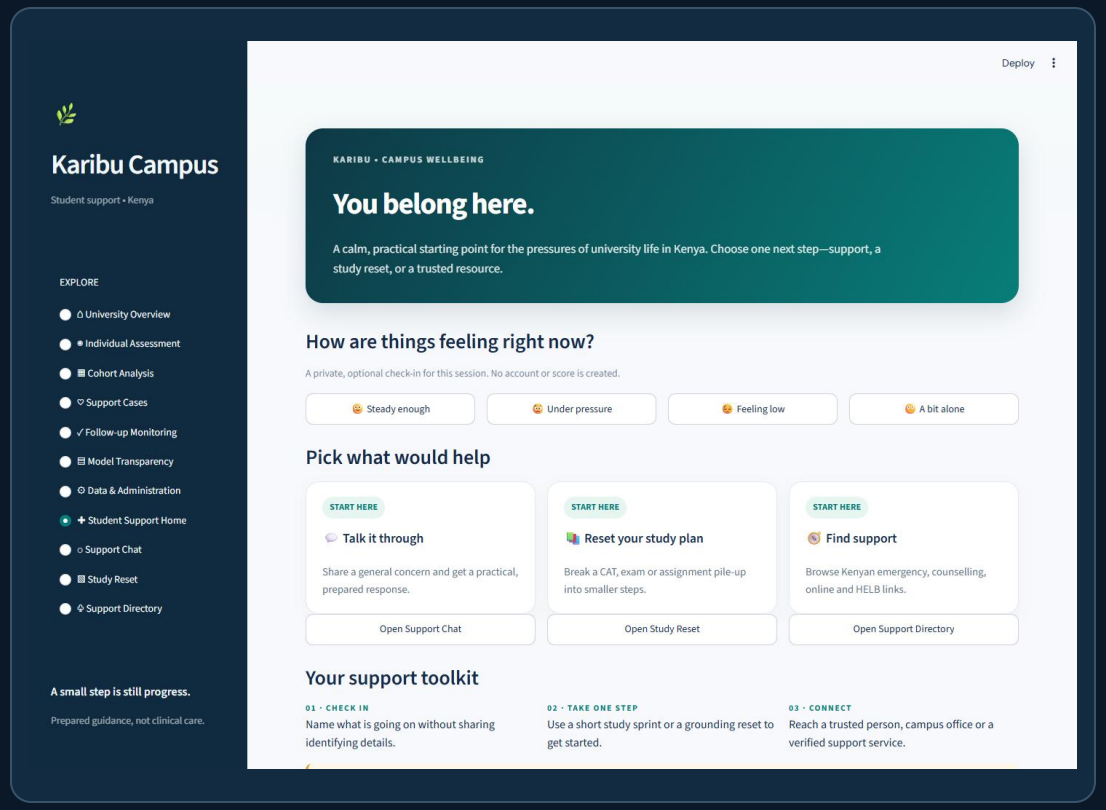
Academic Outcome Analytics

Historical dataset · Classifier comparison
Individual assessment · Cohort analysis

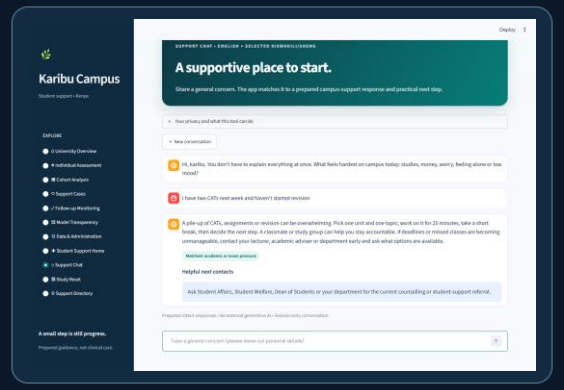
The support hub and outcome model remain separate within the prototype.

Five views preview one navigable application

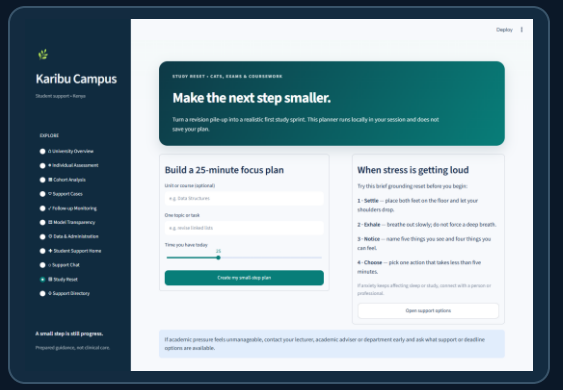
STUDENT SUPPORT HOME



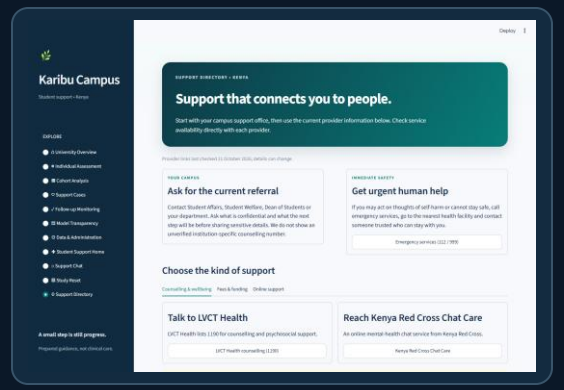
SUPPORT CHAT



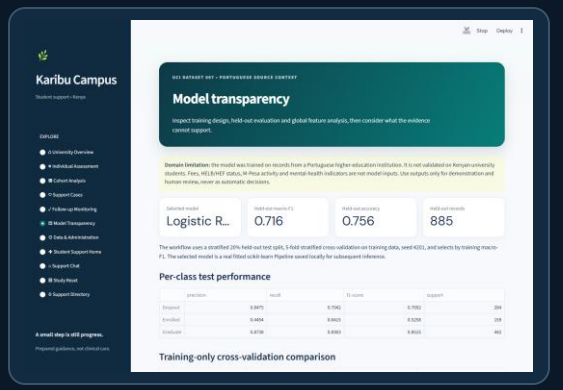
STUDY RESET



SUPPORT DIRECTORY



MODEL TRANSPARENCY



The sidebar connects the student support journey with separate analytics and administration views.

UCI 697 powers the outcome modelling component

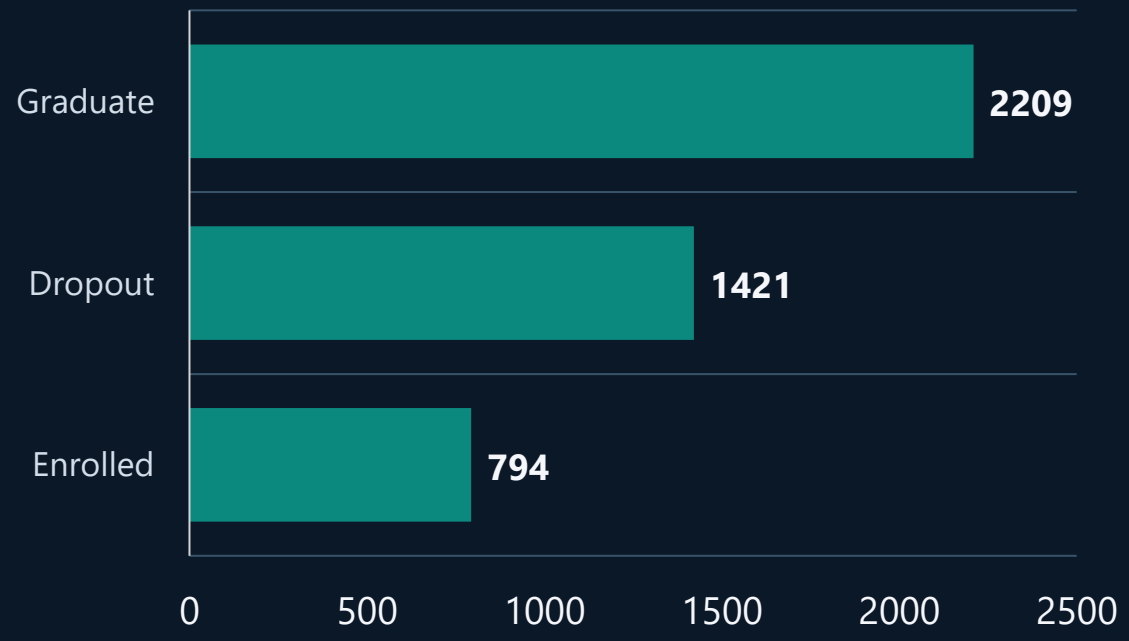
Portuguese higher-education records

4,424 records | 36 predictors | 1 target

FEATURE CATEGORIES IN THE SOURCE DATA

- Demographic information
- Educational background and admission
- Financial indicators
- Semester enrolment and performance

RECORDED TARGET CLASSES



Historical data from Portugal; the model has not been validated on Kenyan university students.

A reproducible workflow, with class-level limits

Stratified 80:20 split · preprocessing inside the scikit-learn Pipeline · select on training CV



0.705
Mean CV macro-F1

0.716
Held-out macro-F1

75.6%
Held-out accuracy

0.526
Enrolled class F1

CV standard deviation: 0.030

Enrolled is the weakest class. All scores describe a held-out split of Portuguese data, not Kenyan validation.

What we will demonstrate

A short route from student support to model-backed analytics

STUDENT SUPPORT

1. Student Support Home
2. Support Chat: CAT or HELB example
3. Study Reset planner
4. Support Directory

DATA AND ANALYTICS

5. Data & Administration
6. Prepare the dataset and train the model
7. Individual Assessment + Cohort Analysis

LIVE APP

https://karibu-campus.onrender.com

Let’s explore Karibu Campus.